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Moisture barrier for metal insulator metal capacitors and integrated circuit having the same

Abstract

A semiconductor die includes a substrate layer, one or more metal layers disposed over the substrate layer, and one or more polyimide layers disposed over the substrate. Each polyimide layer can have a trench that separates a first portion of the polyimide layer that is adjacent a metal layer from a second portion of the polyimide layer. The trench(es) can be formed by etching the polyimide layer(s). A topcoat insulation layer can be disposed over the polyimide layers, a portion of the topcoat insulation layer disposed over the trench(es) define a moat. The topcoat insulation layer is impervious to moisture and the moat inhibits moisture from traveling along the one or more polymer interlevel dielectric layers from the second portion to the first portion.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS (1) Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application, including U.S. Provisional Patent Application No. 62/859,949, filed Jun. 11, 2019 and titled “MOISTURE BARRIER FOR METAL INSULATOR METAL CAPACITORS AND INTEGRATED CIRCUITS HAVING THE SAME,” and U.S. Provisional Patent Application No. 62/860,016, filed Jun. 11, 2019 and titled “METHOD OF MAKING A MOISTURE BARRIER FOR METAL INSULATOR METAL CAPACITORS IN INTEGRATED CIRCUITS,” are hereby incorporated by reference under 37 CFR 1.57.

BACKGROUND

Field

(1) The present disclosure is directed to a moisture barrier for use in an integrated circuit, and more particularly to a moisture barrier for use in integrated circuits that use a polyimide interlayer insulator.

Description of the Related Art

(2) Some semiconductor integrated circuits (ICs) use polyimide as an interlayer insulator. However, during operation under ambient conditions, as well as during high accelerated stress testing (HAST) test conditions (high temperature, high pressure, high moisture conditions), polyimide acts as a wick to draw moisture from outside and transport it to sensitive elements on the ICs, such as capacitors. Such moisture primarily travels from the die edges, bond pads and other cracks in the top coat of the ICs to sensitive elements. Once the moisture reaches silicon nitride dielectric material, the insulating/dielectric properties weaken as the moisture gets in, causing current to pass due to the presence of electric field across the capacitor. The moisture causes the breakdown strength of these sensitive elements to drop and thus lead to failure (e.g. a short of the capacitors), such as temperature, humidity and bias voltage (THB) failures.

SUMMARY OF CERTAIN INVENTIVE ASPECTS

(3) Accordingly, there is a need for an improved manner to avoid moisture from reaching sensitive elements (e.g., capacitors), such as in semiconductor integrated circuits that use polyimide as an interlayer insulator.

(4) In accordance with one aspect of the invention, a packaged module is provided with a moat (e.g., a no polyimide zone) that provides a moisture barrier. In one implementation, the moat is provided around one or more capacitors (e.g., provided around each capacitor). In another implementation, the moat is provided around the die.

(5) In accordance with one aspect of the disclosure, a semiconductor die is provided. The semiconductor die can include a substrate layer, and one or more metal layers that are disposed over the substrate layer. The die can also include one or more polymer interlevel dielectric layers that are disposed over the substrate, each polymer interlevel dielectric layer has a trench that separates a first portion of the interlevel dielectric layer that is adjacent at least one of the one or more metal layers from a second portion of the interlevel dielectric layer. The die can also include a topcoat insulation layer that is disposed over the one or more polymer interlevel dielectric layers and defining a top surface of the die. A portion of the topcoat insulation layer can be disposed over the trench in at least one of the polymer interlevel dielectric layers to define a moat. The topcoat insulation layer is impervious to moisture and the moat inhibits moisture from traveling along the

one or more polymer interlevel dielectric layers from the second portion to the first portion.

(6) In an embodiment, the one or more polymer interlevel dielectric layers are one or more polyimide layers.

(7) In an embodiment, the one or more metal layers are one or more electrodes of a capacitor.

(8) In an embodiment, the semiconductor die further includes one or more insulation layers that are disposed at least partially over at least one of the one or more metal layers.

(9) In an embodiment, the trench in each of the one or more polymer interlevel dielectric layers tapers toward the substrate layer.

(10) In an embodiment, the one or more polymer interlevel dielectric layers are two layers. The trench in each of the two polymer interlevel dielectric layers are stepped relative to each other.

(11) In an embodiment, the moat circumscribes a capacitor of the semiconductor die defined by at least one of the one or more metal layers.

(12) In an embodiment, the moat circumscribes a plurality of electronic components on the semiconductor die.

(13) In accordance with another aspect of the disclosure, a radiofrequency module is provided. The module can include a printed circuit board. The module can also include a semiconductor die mounted on the printed circuit board. The semiconductor die includes a substrate layer, and one or more metal layers disposed over the substrate layer. The die also includes one or more polymer interlevel dielectric layers disposed over the substrate. Each polymer interlevel dielectric layer has a trench that separates a first portion adjacent at least one of the one or more metal layers from a second portion. The die also includes a topcoat insulation layer that is disposed over the one or more polymer interlevel dielectric layers and defining a top surface of the die. A portion of the topcoat insulation layer is disposed over the trench in at least one of the polymer interlevel dielectric layers to define a moat. The topcoat insulation layer is impervious to moisture, and the moat inhibits moisture from traveling along the one or more polymer interlevel dielectric layers from the second portion to the first portion.

(14) In an embodiment, the one or more polymer interlevel dielectric layers are one or more polyimide layers.

(15) In an embodiment, the one or more metal layers are one or more electrodes of a capacitor.

(16) In an embodiment, the radiofrequency module further includes one or more insulation layers that are disposed at least partially over at least one of the one or more metal layers.

(17) In an embodiment, the trench in each of the one or more polymer interlevel dielectric layers tapers toward the substrate layer.

(18) In an embodiment, the one or more polymer interlevel dielectric layers are two layers. The trench in each of the two polymer interlevel dielectric layers can be stepped relative to each other.

(19) In an embodiment, the moat is arranged around a capacitor of the semiconductor die.

(20) In an embodiment, the moat is arranged around a plurality of electronic components on the semiconductor die.

(21) In accordance with another aspect of the disclosure, a wireless mobile device is provided. The wireless mobile device can include one or more antennas, and a front end system that communicates with the one or more antennas. The wireless mobile device can also include a semiconductor die. The die includes a substrate layer, and one or more metal layers disposed over the substrate layer. The die also includes one or more polymer interlevel dielectric layers that is disposed over the substrate. Each polymer interlevel dielectric layer has a trench that separates a first portion adjacent at least one of the one or more metal layers from a second portion. The die also includes a topcoat insulation layer that is disposed over the one or more polymer interlevel dielectric layers and defining a top surface of the die. A portion of the topcoat insulation layer is disposed over the trench in at least one of the polymer interlevel dielectric layers to define a moat. The topcoat insulation layer is impervious to moisture, and the moat inhibits moisture from traveling along the one or more polymer interlevel dielectric layers from the second portion to the

first portion.

(22) In an embodiment, the one or more polymer interlevel dielectric layers are one or more polyimide layers.

(23) In an embodiment, the one or more metal layers are one or more electrodes of a capacitor.

(24) In an embodiment, the wireless mobile device further includes one or more insulation layers that are disposed at least partially over at least one of the one or more metal layers.

(25) In an embodiment, the trench in each of the one or more polymer interlevel dielectric layers tapers toward the substrate layer.

(26) In an embodiment, the one or more polymer interlevel dielectric layers are two layers. The trench in each of the two polymer interlevel dielectric layers can be stepped relative to each other.

(27) In an embodiment, the moat is arranged around a capacitor of the semiconductor die.

(28) In an embodiment, the moat is arranged around a plurality of electronic components on the semiconductor die.

(29) In accordance with another aspect of the disclosure, a method of making a semiconductor die is provided. The method can include forming or providing a substrate layer, forming or applying one or more metal layers over the substrate layer, and forming or applying one or more polymer interlevel dielectric layers over the substrate layer. At least a first portion of the one or more polymer interlevel dielectric layers is adjacent at least one of the one or more metal layers. The method can also include forming a trench in the one or more polymer interlevel dielectric layers to separate the first portion from a second portion of the one or more polymer interlevel dielectric layers. The method can also include forming or applying a topcoat insulation layer over the one or more polymer interlevel dielectric layers. A portion of the topcoat insulation layer is disposed over the trench in at least one of the polymer interlevel dielectric layers to define a moat. The topcoat insulation layer being impervious to moisture. The moat inhibits moisture from traveling along the one or more polymer interlevel dielectric layers from the second portion to the first portion.

(30) In an embodiment, the one or more polymer interlevel dielectric layers are one or more polyimide interlevel dielectric layers.

(31) In an embodiment, the one or more polymer interlevel dielectric layers are two polymer interlevel dielectric layers. The trench can be formed in each of the two polymer interlevel dielectric layers in a stepped manner relative to each other.

(32) In an embodiment, forming the trench includes etching the one or more polymer interlevel dielectric layers to form the trench.

(33) In an embodiment, the one or more metal layers are one or more electrodes of a capacitor.

(34) In an embodiment, the method further includes forming or applying one or more insulation layers at least partially over at least one of the one or more metal layers.

(35) In an embodiment, the moat is arranged around a capacitor of the semiconductor die.

(36) In an embodiment, the moat is arranged around a plurality of electronic components on the semiconductor die.

(37) In accordance with another aspect of the disclosure, a method of making a radiofrequency module is provided. The method can include forming or providing a printed circuit board that includes a substrate layer, forming or providing a semiconductor die, and mounting the semiconductor die on the printed circuit board. Forming or providing the semiconductor die includes forming or providing a substrate layer, forming or applying one or more metal layers over the substrate layer, and forming or applying one or more polymer interlevel dielectric layers over the substrate layer. At least a first portion of the one or more polymer interlevel dielectric layers is adjacent at least one of the one or more metal layers. Forming or providing the die further includes forming a trench in the one or more polymer interlevel dielectric layers to separate the first portion from a second portion of the one or more polymer interlevel dielectric layers, and forming or applying a topcoat insulation layer over the one or more polymer interlevel dielectric layers. A portion of the topcoat insulation layer is disposed over the trench in at least one of the polymer

interlevel dielectric layers to define a moat. The topcoat insulation layer is impervious to moisture, and the moat inhibits moisture from traveling along the one or more polymer interlevel dielectric layers from the second portion to the first portion.

(38) In an embodiment, the one or more polymer interlevel dielectric layers are one or more polyimide interlevel dielectric layers.

(39) In an embodiment, the one or more polymer interlevel dielectric layers are two polymer interlevel dielectric layers. The trench can be formed in each of the two polymer interlevel dielectric layers in a stepped manner relative to each other.

(40) In an embodiment, forming the trench includes etching the one or more polymer interlevel dielectric layers to form the trench.

(41) In an embodiment, the one or more metal layers are one or more electrodes of a capacitor.

(42) In an embodiment, the method can further include forming or applying one or more insulation layers at least partially over at least one of the one or more metal layers.

(43) In an embodiment, forming the trench includes tapering the trench toward the substrate layer.

(44) In an embodiment, the moat is arranged around a capacitor of the semiconductor die.

(45) In an embodiment, the moat is arranged around a plurality of electronic components on the semiconductor die.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of this disclosure will now be described, by way of non-limiting example, with reference to the accompanying drawings.

(2) FIG. 1 is a schematic diagram of one embodiment of a wireless device.

(3) FIG. 2A is a schematic diagram of one embodiment of a packaged module.

(4) FIG. 2B is a schematic diagram of a cross-section of the packaged module of FIG. 2A taken along the lines 2B-2B.

(5) FIG. 3A is a schematic top view of a die with a capacitor and moat without any connections over the moat.

(6) FIG. 3B is a cross-sectional side view of the die in FIG. 3A including the moat.

(7) FIG. 4A is a schematic top view of a die with a capacitor and moat with a connection over the moat.

(8) FIG. 4B is a cross-sectional side view of the die in FIG. 4A including the moat.

(9) FIG. 5A is a schematic top view of a die with a capacitor and moat with a connection over the moat.

(10) FIG. 5B is a cross-sectional side view of the die in FIG. 5A including the moat.

(11) FIG. 6A is a schematic top view of a die with a capacitor and moat with a connection over the moat.

(12) FIG. 6B is a cross-sectional side view of the die in FIG. 6A including the moat.

(13) FIG. 7A is a schematic top view of a die with a capacitor and moat with a connection over the moat.

(14) FIG. 7B is a cross-sectional side view of the die in FIG. 7A including the moat and without a connection over the moat.

(15) FIG. 7C is a cross-sectional side view of the die in FIG. 7A including the moat and without a connection over the moat.

(16) FIG. 7D is a cross-sectional side view of the die in FIG. 7A including the moat and without a connection over the moat.

(17) FIG. 8 is a schematic top view of a die with a moat formed around a portion of the die.

(18) FIG. 9 is a block diagram of a method for manufacturing a die with a moat in one or more

polymer layers.

DETAILED DESCRIPTION OF CERTAIN EMBODIMENTS

(19) The following detailed description of certain embodiments presents various descriptions of specific embodiments. However, the innovations described herein can be embodied in a multitude of different ways, for example, as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals can indicate identical or functionally similar elements. It will be understood that elements illustrated in the figures are not necessarily drawn to scale. Moreover, it will be understood that certain embodiments can include more elements than illustrated in a drawing and/or a subset of the elements illustrated in a drawing. Further, some embodiments can incorporate any suitable combination of features from two or more drawings.

(20) The International Telecommunication Union (ITU) is a specialized agency of the United Nations (UN) responsible for global issues concerning information and communication technologies, including the shared global use of radio spectrum.

(21) The 3rd Generation Partnership Project (3GPP) is a collaboration between groups of telecommunications standard bodies across the world, such as the Association of Radio Industries and Businesses (ARIB), the Telecommunications Technology Committee (TTC), the China Communications Standards Association (CCSA), the Alliance for Telecommunications Industry Solutions (ATIS), the Telecommunications Technology Association (TTA), the European Telecommunications Standards Institute (ETSI), and the Telecommunications Standards Development Society, India (TSDSI).

(22) Working within the scope of the ITU, 3GPP develops and maintains technical specifications for a variety of mobile communication technologies, including, for example, second generation (2G) technology (for instance, Global System for Mobile Communications (GSM) and Enhanced Data Rates for GSM Evolution (EDGE)), third generation (3G) technology (for instance, Universal Mobile Telecommunications System (UMTS) and High Speed Packet Access (HSPA)), and fourth generation (4G) technology (for instance, Long Term Evolution (LTE) and LTE-Advanced).

(23) The technical specifications controlled by 3GPP can be expanded and revised by specification releases, which can span multiple years and specify a breadth of new features and evolutions.

(24) In one example, 3GPP introduced carrier aggregation (CA) for LTE in Release 10. Although initially introduced with two downlink carriers, 3GPP expanded carrier aggregation in Release 14 to include up to five downlink carriers and up to three uplink carriers. Other examples of new features and evolutions provided by 3GPP releases include, but are not limited to, License Assisted Access (LAA), enhanced LAA (eLAA), Narrowband Internet-of-Things (NB-IOT), Vehicle-to-Everything (V2X), and High Power User Equipment (HPUE).

(25) 3GPP introduced Phase 1 of fifth generation (5G) technology in Release 15 and plans to introduce Phase 2 of 5G technology in Release 16 (targeted for 2019). Subsequent 3GPP releases will further evolve and expand 5G technology. 5G technology is also referred to herein as 5G New Radio (NR).

(26) 5G NR supports or plans to support a variety of features, such as communications over millimeter wave spectrum, beam forming capability, high spectral efficiency waveforms, low latency communications, multiple radio numerology, and/or non-orthogonal multiple access (NOMA). Although such RF functionalities offer flexibility to networks and enhance user data rates, supporting such features can pose a number of technical challenges.

(27) The teachings herein are applicable to a wide variety of communication systems, including, but not limited to, analog devices, radiofrequency devices, and communication systems using advanced cellular technologies, such as LTE-Advanced, LTE-Advanced Pro, and/or 5G NR.

(28) FIG. 1 is a schematic diagram of one embodiment of a wireless device **100**. The wireless device **100** can be, for example but not limited to, a portable telecommunication device such as a mobile cellular-type telephone. The wireless device **100** can include one or more of a baseband system **101**, a transceiver **102**, a front end system **103**, one or more antennas **104**, a power

management system **105**, a memory **106**, a user interface **107**, a battery **108** (e.g., direct current (DC) battery). Other additional components, such as a speaker, display and keyboard can optionally be connected to the baseband system **101**. The battery **108** can provide power to the wireless device **100**.

(29) It should be noted that, for simplicity, only certain components of the wireless device **100** are illustrated herein. The control signals provided by the baseband system **101** control the various components within the wireless device **100**. Further, the function of the transceiver **102** can be integrated into separate transmitter and receiver components.

(30) The wireless device **100** can be used communicate using a wide variety of communications technologies, including, but not limited to, 2G, 3G, 4G (including LTE, LTE-Advanced, and LTE-Advanced Pro), 5G NR, WLAN (for instance, Wi-Fi), WPAN (for instance, Bluetooth and ZigBee), WMAN (for instance, WiMax), and/or GPS technologies.

(31) The transceiver **102** generates RF signals for transmission and processes incoming RF signals received from the antennas **104**. It will be understood that various functionalities associated with the transmission and receiving of RF signals can be achieved by one or more components that are collectively represented in FIG. **1** as the transceiver **102**. In one example, separate components (for instance, separate circuits or dies) can be provided for handling certain types of RF signals.

(32) The front end system **103** aids in conditioning signals transmitted to and/or received from the antennas **104**. In the illustrated embodiment, the front end system **103** includes one or more power amplifiers (PAs) **111**, low noise amplifiers (LNAs) **112**, filters **113**, switches **114**, duplexers **115** and capacitors **300**. However, other implementations are possible.

(33) For example, the front end system **103** can provide a number of functionalities, including, but not limited to, amplifying signals for transmission, amplifying received signals, filtering signals, switching between different bands, switching between different power modes, switching between transmission and receiving modes, duplexing of signals, multiplexing of signals (for instance, diplexing or triplexing), or some combination thereof.

(34) In certain implementations, the wireless device **100** supports carrier aggregation, thereby providing flexibility to increase peak data rates. Carrier aggregation can be used for both Frequency Division Duplexing (FDD) and Time Division Duplexing (TDD), and may be used to aggregate a plurality of carriers or channels. Carrier aggregation includes contiguous aggregation, in which contiguous carriers within the same operating frequency band are aggregated. Carrier aggregation can also be non-contiguous, and can include carriers separated in frequency within a common band or in different bands.

(35) The antennas **104** can include antennas used for a wide variety of types of communications. For example, the antennas **104** can include antennas for transmitting and/or receiving signals associated with a wide variety of frequencies and communications standards.

(36) In certain implementations, the antennas **104** support MIMO communications and/or switched diversity communications. For example, MIMO communications use multiple antennas for communicating multiple data streams over a single radio frequency channel. MIMO communications benefit from higher signal to noise ratio, improved coding, and/or reduced signal interference due to spatial multiplexing differences of the radio environment. Switched diversity refers to communications in which a particular antenna is selected for operation at a particular time. For example, a switch can be used to select a particular antenna from a group of antennas based on a variety of factors, such as an observed bit error rate and/or a signal strength indicator.

(37) The wireless device **100** can operate with beamforming in certain implementations. For example, the front end system **103** can include phase shifters having variable phase controlled by the transceiver **102**. Additionally, the phase shifters are controlled to provide beam formation and directivity for transmission and/or reception of signals using the antennas **104**. For example, in the context of signal transmission, the phases of the transmit signals provided to the antennas **104** are controlled such that radiated signals from the antennas **104** combine using constructive and

destructive interference to generate an aggregate transmit signal exhibiting beam-like qualities with more signal strength propagating in a given direction. In the context of signal reception, the phases are controlled such that more signal energy is received when the signal is arriving to the antennas **104** from a particular direction. In certain implementations, the antennas **104** include one or more arrays of antenna elements to enhance beamforming.

(38) The baseband system **101** is coupled to the user interface **107** to facilitate processing of various user input and output (I/O), such as voice and data. The baseband system **101** provides the transceiver **102** with digital representations of transmit signals, which the transceiver **102** processes to generate RF signals for transmission. The baseband system **101** also processes digital representations of received signals provided by the transceiver **102**. As shown in FIG. 1, the baseband system **101** is coupled to the memory **106** of facilitate operation of the wireless device **100**.

(39) The memory **106** can be used for a wide variety of purposes, such as storing data and/or instructions to facilitate the operation of the wireless device **100** and/or to provide storage of user information.

(40) The power management system **105** provides a number of power management functions of the wireless device **100**. In certain implementations, the power management system **105** includes a PA supply control circuit that controls the supply voltages of the power amplifiers **111**. For example, the power management system **105** can be configured to change the supply voltage(s) provided to one or more of the power amplifiers **111** to improve efficiency, such as power added efficiency (PAE).

(41) As shown in FIG. 1, the power management system **105** receives a battery voltage from the battery **108**. The battery **108** can be any suitable battery for use in the wireless device **100**, including, for example, a lithium-ion battery.

(42) FIG. 2A is a schematic diagram of one embodiment of a packaged module **200**. FIG. 2B is a schematic diagram of a cross-section of the packaged module **200** of FIG. 2A taken along the lines 2B-2B.

(43) The packaged module **200** includes radio frequency components **201**, a semiconductor die **202**, surface mount devices **203**, wirebonds **208**, a package substrate **230**, an encapsulation structure **240**, and one or more capacitors **300**. The package substrate **230** includes pads **206** formed from conductors disposed therein. Additionally, the semiconductor die **202** includes pins or pads **204**, and the wirebonds **208** have been used to connect the pads **204** of the die **202** to the pads **206** of the package substrate **220**.

(44) The semiconductor die **202** includes a power amplifier **245**, which can be implemented in accordance with one or more features disclosed herein.

(45) The package substrate **230** can be configured to receive a plurality of components such as radio frequency components **201**, the semiconductor die **202** and the surface mount devices **203**, which can include, for example, surface mount capacitors and/or inductors. In one implementation, the radio frequency components **201** include integrated passive devices (IPDs).

(46) As shown in FIG. 2B, the packaged module **200** is shown to include a plurality of contact pads **232** disposed on the side of the packaged module **200** opposite the side used to mount the semiconductor die **202**. Configuring the packaged module **200** in this manner can aid in connecting the packaged module **200** to a circuit board, such as a phone board of a mobile device. The example contact pads **232** can be configured to provide radio frequency signals, bias signals, and/or power (for example, a power supply voltage and ground) to the semiconductor die **202** and/or other components. As shown in FIG. 2B, the electrical connections between the contact pads **232** and the semiconductor die **202** can be facilitated by connections **233** through the package substrate **230**.

The connections **233** can represent electrical paths formed through the package substrate **220**, such as connections associated with vias and conductors of a multilayer laminated package substrate.

(47) In some embodiments, the packaged module **200** can also include one or more packaging

structures to, for example, provide protection and/or facilitate handling. Such a packaging structure can include overmold or encapsulation structure **240** formed over the package substrate **230** and the components and die(s) disposed thereon.

(48) It will be understood that although the packaged module **200** is described in the context of electrical connections based on wirebonds, one or more features of the present disclosure can also be implemented in other packaging configurations, including, for example, flip-chip configurations.

(49) Moisture Barrier

(50) FIGS. **3A-3B** schematically illustrate a die **500A** with a capacitor **300A** and a moat **400A**. FIG. **3A** shows only a portion of the moat **400A**. In one implementation, the moat **400A** can extend about (e.g., completely circumscribe or surround) the capacitor **300A** to provide a moisture barrier (e.g., a local moat) about the capacitor **300A**. In another implementation, the moat **400A** can extend about (e.g., completely circumscribe or surround) one or more (e.g., a plurality of) components (e.g., a plurality of capacitors) to provide a moisture barrier about the one or more (e.g., a plurality of) components. In still another implementation, the moat **400A** can extend about (e.g., completely circumscribe or surround) at least a portion of a die, such as the die **500A**, to provide a moisture barrier from defects on one or more edges of the die.

(51) FIG. **3B** shows a cross-section of the die **500A** along line **3B-3B** in FIG. **3A**. The die **500A** includes a substrate layer **510A**. Optionally, the substrate layer **510A** can be made of gallium arsenide (GaAs). An insulation layer **520A** is disposed over (e.g., directly over, adjacent) the substrate layer **510A**. Optionally, the insulation layer **520A** can be made of silicon nitride (SiN).

(52) A first metal layer **530A** (M1) can be disposed over (e.g., directly over, adjacent) at least a portion of the insulation layer **520A**. The first metal layer **530A** can define a first electrode and extend along a length smaller than a length of the insulation layer **520A**. A second insulation layer **540A** can be disposed over (e.g., directly over, adjacent) the first metal layer **530A** and disposed over the insulation layer **520A**. The second insulation layer **540A** can optionally be made of silicon nitride.

(53) A second metal layer **550A** (M2) can be disposed over at least a portion of (e.g., less than an entire length of) the second insulation layer **540A**. The second metal layer **550A** can define a second electrode and extend along a length smaller than a length of the second insulation layer **540A**. The second metal layer **550A** can be shorter in length than the first metal layer **530A**. A third insulation layer **560A** can be disposed over at least a portion of (e.g., less than an entire length of) the second metal layer **550A** and can extend over at least a portion of the second insulation layer **540A**. A first interlayer insulator layer **570A** (V1) is disposed over (e.g., directly over, adjacent) the third insulation layer **560A**. The first interlayer insulator layer **570A** can be made of polyimide. However, the first interlayer insulator layer **570A** can be made of other suitable polymer materials.

(54) A third metal layer **580A** (M3) can be disposed over (e.g., directly over, adjacent) at least a portion of (e.g., less than an entire length of) the second metal layer **550A**, as well as over at least a portion of (e.g., less than an entire length of) the first interlayer insulator layer **570A**. the third metal layer **580A** can define a third electrode and extend along a length approximately equal to the length of the first metal layer **530A**. A second interlayer insulator layer **590A** (V2) can be disposed over (e.g., directly over, adjacent) the third metal layer **580A** and over (e.g., directly over, adjacent) at least a portion of the first interlayer insulator layer **570A**. the second interlayer insulator layer **590A** can be made of polyimide. However, the second interlayer insulator layer **590A** can be made of other suitable polymer materials. A topcoat insulation layer (or fourth insulation layer) **600A** can be disposed over (e.g. directly over, adjacent) at least a portion of the second interlayer insulator layer **590A** and at last a portion of the first interlayer insulator layer **570A**. The topcoat insulation layer **600A** can optionally be made of silicon nitride, and is impervious to moisture, thereby providing a moisture barrier.

(55) The capacitor **300A** can be defined by one or more of the first, second and third metal layers **530A**, **550A**, **580A**. The moat **400A** can be defined in the die **500A** by a trench portion **420A**

defined in the first interlayer insulator layer **570A** and a trench portion **440A** defined in the second interlayer insulator layer **590A** to define a trench **630A** in the topcoat insulation layer **600A** that extends over the trench portions **420A**, **440A**. The trench portion **440A** can be wider than the trench portion **420A**, both having a tapered shape. In one implementation, as shown in FIG. 3A where all dimensions are in microns, the trench portion **440A** can be approximately 6 microns wide at its widest point, and the trench portion **420A** can be approximately 4 microns wide at its widest point. However, the width of the trench portions **420A**, **440A** can have other suitable dimensions. An inner surface of the trenches **420A**, **440A** optionally are offset to define a stepped tapered shape for the trench **630A**. As discussed above, the topcoat insulation layer **600A** extends over (e.g., adjacent) the trench portions **420A**, **440A** in the first and second interlayer insulator layers **570A**, **590A**. As shown in FIG. 3B, the trench **630A** of the topcoat insulation layer **600A** can also extend within a trench portion **620A** defined in the first and second insulation layers **520A**, **540A** to a location over (e.g., adjacent) a portion of the substrate **510A**. Therefore, the topcoat insulation layer **600A** advantageously provides a moisture barrier (e.g., with the trench **630A**) between different portions of the first interlayer insulator layer **570A** and between different portions of the second interlayer insulator layer **590A**, thereby inhibiting (e.g., preventing) moisture from flowing from one portion (e.g., right side of moat **400A** in FIG. 3B) of the first and second interlayer insulator layers **570A**, **590A** to another portion (e.g., left side of moat **400A** in FIG. 3B) of the first and second interlayer insulator layers **570A**, **590A**. Such a moisture barrier advantageously inhibits (e.g., prevents) damage to the capacitor **300A** from exposure to moisture that may travel along one or both of the first and second interlayer insulator layers **570A**, **590A** (e.g., from defects on a top portion or an edge portion of the die **500A**). As shown in FIGS. 3A-3B, there is no electrical connection across the moat **400A**. That is, none of the first, second or third metal layers **530A**, **550A**, **580A** extend across the moat **400A** or trench **630A**. Further, as shown in FIG. 3A, in one implementation, the moat **400A** can be spaced approximately 3 microns from the capacitor **300A**. However, the spacing between the moat **400A** and the capacitor **300A** can have other suitable dimensions.

(56) FIGS. 4A-4B schematically illustrate a top view and a cross-sectional view along line 4B-4B in FIG. 4A, respectively, of a die **500B** with a capacitor **300B** and a moat **400B**. Some of the features of the die **500B**, capacitor **300B** and moat **400B** are similar to features of the die **500A**, capacitor **300A** and moat **400A** in FIGS. 3A-3B. Thus, references numerals used to designate the various components of the die **500B**, capacitor **300B** and moat **400B** are identical to those used for identifying the corresponding components of the die **500A**, capacitor **300A** and moat **400A** in FIGS. 3A-3B, except that a "B" instead of an "A" has been added to the numerical identifier. Therefore, the structure and description for the various features of the die **500A**, capacitor **300A** and moat **400A** in FIGS. 3A-3B are understood to also apply to the corresponding features of the die **500B**, capacitor **300B** and moat **400B** in FIGS. 4A-4B, except as described below.

(57) The die **500B**, capacitor **300B** and moat **400B** differ from the die **500A**, capacitor **300A** and moat **400A** in that the first metal layer **530B** extends across the moat **400B**, thereby allowing an electrical communication across the moat **400B**. The topcoat insulation layer (e.g., fourth insulation layer) **600B** extends over the trench portions **420B**, **440B** in the first and second interlayer insulator layers **570B**, **590B** and over the trench portion **620B** in the first and second insulation layers **520B**, **540B** to a location over (e.g., adjacent) a portion of the first metal layer **530B**, thereby defining a trench **630B** in the topcoat insulation layer **600B** that extends over the trench portions **420B**, **440B**. Therefore, the topcoat insulation layer **600B** advantageously provides a moisture barrier between different portions of the first interlayer insulator layer **570B** and between different portions of the second interlayer insulator layer **590B**. This moisture barrier inhibits (e.g., prevents) moisture from flowing from one portion (e.g., right side of moat **400B** in FIG. 4B) of the first and second interlayer insulator layers **570B**, **590B** to another portion (e.g., left side of moat **400B** in FIG. 4B) of the first and second interlayer insulator layers **570B**, **590B**, thereby advantageously inhibiting

(e.g., preventing) damage to the capacitor **300B** from exposure to moisture that may travel along one or both of the first and second interlayer insulator layers **570B**, **590B** (e.g., from defects on a top portion or an edge portion of the die **500B**).

(58) FIGS. **5A-5B** schematically illustrate a top view and a cross-sectional view along line **5B-5B** in FIG. **5A**, respectively, of a die **500C** with a capacitor **300C** and a moat **400C**. Some of the features of the die **500C**, capacitor **300C** and moat **400C** are similar to features of the die **500A**, capacitor **300A** and moat **400A** in FIGS. **3A-3B**. Thus, references numerals used to designate the various components of the die **500C**, capacitor **300C** and moat **400C** are identical to those used for identifying the corresponding components of the die **500A**, capacitor **300A** and moat **400A** in FIGS. **3A-3B**, except that a “C” instead of an “A” has been added to the numerical identifier. Therefore, the structure and description for the various features of the die **500A**, capacitor **300A** and moat **400A** in FIGS. **3A-3B** are understood to also apply to the corresponding features of the die **500C**, capacitor **300C** and moat **400C** in FIGS. **5A-5B**, except as described below.

(59) The die **500C**, capacitor **300C** and moat **400C** differ from the die **500A**, capacitor **300A** and moat **400A** in that the third metal layer **580C** extends across the moat **400C**, thereby allowing an electrical communication across the moat **400C**. The first metal layer **530C** also provides a footer or pedestal **535C** that supports at least a portion of the third metal layer **580C**, as further discussed below.

(60) The topcoat insulation layer (e.g., fourth insulation layer) **600C** extends over the trench portion **440C** in the second interlayer insulator layer **590C** and over (e.g., directly over, adjacent) at least a portion of the third metal layer **580C**. The third metal layer **580C** defines a trench portion **585C** (e.g., tapered trench portion). The portion of the topcoat insulation layer **600C** that extends over the trench portion **440C** in the second interlayer insulator layer **590C** and the trench portion **585C** of the third metal layer **580C** defines a trench **630C** (e.g., tapered or tapered step trench portion).

(61) The trench portion **585C** of the third metal layer **580C** extends over (e.g., directly over, adjacent) the trench portion **420C** in the first interlayer insulator layer **570C** to a location over (e.g., directly over, adjacent) the footer or pedestal **535C**. The pedestal **535A** advantageously allows the trench portion **585C** in the third metal layer **580C** and the trench **630C** in the topcoat insulation layer **600C** to have a depth that inhibits (e.g., prevents) failure or damage to one or more layers of the die **500C**.

(62) Therefore, the topcoat insulation layer **600C** advantageously provides a moisture barrier between different portions of the second interlayer insulator layer **590C**, and the third metal layer **580C** provide a barrier between different portions of the first interlayer insulator layer **570C**. This moisture barrier inhibits (e.g., prevents) moisture from flowing from one portion (e.g., right side of moat **400C** in FIG. **5B**) of the first and second interlayer insulator layers **570C**, **590C** to another portion (e.g., left side of moat **400C** in FIG. **5B**) of the first and second interlayer insulator layers **570C**, **590C**, thereby advantageously inhibiting (e.g., preventing) damage to the capacitor **300C** from exposure to moisture that may travel along one or both of the first and second interlayer insulator layers **570C**, **590C** (e.g., from defects on a top portion or an edge portion of the die **500C**).

(63) FIGS. **6A-6B** schematically illustrate a top view and a cross-sectional view along line **6B-6B** in FIG. **6A**, respectively, of a die **500D** with a capacitor **300D** and a moat **400D**. Some of the features of the die **500D**, capacitor **300D** and moat **400D** are similar to features of the die **500C**, capacitor **300C** and moat **400C** in FIGS. **5A-5B**. Thus, references numerals used to designate the various components of the die **500D**, capacitor **300D** and moat **400D** are identical to those used for identifying the corresponding components of the die **500C**, capacitor **300C** and moat **400C** in FIGS. **5A-5B**, except that a “D” instead of a “C” has been added to the numerical identifier. Therefore, the structure and description for the various features of the die **500C**, capacitor **300C** and moat **400C** in FIGS. **5A-5B** are understood to also apply to the corresponding features of the

die **500D**, capacitor **300D** and moat **400D** in FIGS. **6A-6B**, except as described below.

(64) The die **500D**, capacitor **300D** and moat **400D** differ from the die **500C**, capacitor **300C** and moat **400C** in that the third metal layer **580D** does not extend across the moat **400D**. Rather, the third metal layer **580D** has a first portion **583D** that extends along a length substantially equal to the length of the first metal layer **530D** and shorter than a length of the length of the second insulation layer **540D**. The third metal layer **580D** also has a second portion **586D** that defines a trench portion **585D** that is supported on (e.g., disposed over, adjacent, in contact with) the footer or pedestal **535D** of the first metal layer **530D**. The trench portion **585D** extends over the trench portion **420D** in the first interlayer insulator layer **570D**.

(65) The die **500D** also includes a fourth metal layer **595D** (M3) that extends across the moat **400D**, thereby allowing an electrical communication across the moat **400D**. The fourth metal layer **595D** extends over (e.g., directly over, adjacent) the second interlayer insulator layer **590D**. The fourth metal layer **595D** has a trench portion **597D** (e.g., tapered trench portion) that extends over the trench portion **440D** in the second interlayer insulator layer **590D** and at least partially extends into the trench portion **585D** of the third metal layer **580D**. The fourth metal layer **595D** can define a fourth electrode.

(66) The topcoat insulation layer (e.g., fourth insulation layer) **600D** extends over (e.g., directly over, adjacent) the fourth metal layer **595D**. The portion of the topcoat insulation layer **600D** that extends over the trench portion **597D** in the fourth metal layer **595D** defines a trench **630D** (e.g., tapered or tapered step trench portion).

(67) The pedestal **535D** advantageously allows the trench portion **585D** in the third metal layer **580D**, the trench portion **597D** in the fourth metal layer **595D**, and the trench **630D** in the topcoat insulation layer **600D** to have a depth that inhibits (e.g., prevents) failure or damage to one or more layers of the die **500D**.

(68) Therefore, the topcoat insulation layer **600D** and fourth metal layer **595D** advantageously provide a moisture barrier between different portions of the second interlayer insulator layer **590D**, and the trench portion **585D** of the third metal layer **580D** provides a barrier between different portions of the first interlayer insulator layer **570D**. This moisture barrier inhibits (e.g., prevents) moisture from flowing from one portion (e.g., right side of moat **400D** in FIG. **6B**) of the first and second interlayer insulator layers **570D**, **590D** to another portion (e.g., left side of moat **400D** in FIG. **6B**) of the first and second interlayer insulator layers **570D**, **590D**, thereby advantageously inhibiting (e.g., preventing) damage to the capacitor **300D** from exposure to moisture that may travel along one or both of the first and second interlayer insulator layers **570D**, **590D** (e.g., from defects on a top portion or an edge portion of the die **500D**).

(69) FIG. **7A** schematically illustrates a top view of a die **500E** with a capacitor **300E** and a moat **400E**. FIG. **7B** shows a cross-sectional view taken along line **7B** in FIG. **7A**. FIG. **7C** shows a cross-sectional view taken along line **7C** in FIG. **7A**. FIG. **7D** shows a cross-sectional view taken along line **7D** in FIG. **7A**. Some of the features of the die **500E**, capacitor **300E** and moat **400E** are similar to features of the die **500C**, capacitor **300C** and moat **400C** in FIGS. **5A-5B**. Other features of the die **500E**, capacitor **300E** and moat **400E** are similar to features of the die **500B**, capacitor **300B** and moat **400B** in FIGS. **4A-4B**. Thus, references numerals used to designate the various components of the die **500E**, capacitor **300E** and moat **400E** are identical to those used for identifying the corresponding components of the die **500B**, **500C**, capacitor **300B**, **300C** and moat **400B**, **400C** in FIGS. **4A-5B**, except that a “E” instead of a “B” or “C” has been added to the numerical identifier. Therefore, the structure and description for the various features of the die **500B**, **500C** capacitor **300B**, **300C** and moat **400B**, **400C** in FIGS. **4A-5B** are understood to also apply to the corresponding features of the die **500E**, capacitor **300E** and moat **400E** in FIGS. **7A-7D**, except as described below.

(70) As shown in FIG. **7A**, the moat **400E** around (e.g., surrounds, circumscribes) the capacitor **300E**, thereby providing a moisture barrier between the capacitor **300E** and area outside the moat

400E. The trench portion **440E** of the moat **400E** extends around (e.g., surrounds, circumscribes) the capacitor **300E**, while the trench portion **420E** of the moat **400E** extends at least partially around (e.g., less than entirely around) the capacitor **300E**.

(71) FIG. 7B shows a cross-sectional view of the die **500E** along line 7B in FIG. 7A, where no electrical connection is provided across the moat **400E**. The cross-section of the die **500E** along line 7B in FIG. 7A is similar to the cross-section of the die **500B** in FIG. 4B, except that the first metal layer **530E** does not extend completely across the die **500E**. The first metal layer **530E** extends across the moat **400E** but has a length smaller than a length of the substrate **510E**. The trench **630E** in the topcoat insulation layer **600E** extends over the trench portions **420E**, **440E** of the moat **400E**. Therefore, the topcoat insulation layer **600E** advantageously provides a moisture barrier between different portions of the first interlayer insulator layer **570E** and between different portions of the second interlayer insulator layer **590E**. This moisture barrier inhibits (e.g., prevents) moisture from flowing across the moat **400E**, thereby advantageously inhibiting (e.g., preventing) damage to the capacitor **300E** from exposure to moisture (e.g., from defects on a top portion or an edge portion of the die **500E**).

(72) FIG. 7C shows a cross-sectional view of the die **500E** along line 7C in FIG. 7A, where an electrical connection is provided across the moat **400E** by the first metal layer **530E**. The cross-section of the die **500E** along line 7C in FIG. 7A is identical to the cross-section of the die **500B** in FIG. 4B, and therefore the description provided above for FIG. 4B also applies to the cross-section of the die **500E** in FIG. 7C.

(73) FIG. 7D shows a cross-sectional view of the die **500E** along line 7D in FIG. 7A, where an electrical connection is provided across the moat **400E** by the third metal layer **580E**. The cross-section of the die **500E** along line 7D in FIG. 7A is similar to the cross-section of the die **500E** along line 7B in FIG. 7A, except that the third metal layer **580E** extends across the moat **400E** and has a length approximately equal to the length of the substrate **510E**. Additionally, there is no trench in the first interlayer insulator layer **570E**. The third metal layer **580E** extends over (e.g., directly over, adjacent, in contact with) the first interlayer insulator layer **570E**. The trench **630E** in the topcoat insulation layer **600E** extends over the trench portion **440E** of the second interlayer insulator layer **590E**. Therefore, the topcoat insulation layer **600E** advantageously provides a moisture barrier between different portions of the second interlayer insulator layer **590E**. This moisture barrier inhibits (e.g., prevents) moisture from flowing across the moat **400E** via the second interlayer insulator layer **590E**, thereby advantageously inhibiting (e.g., preventing) damage to the capacitor **300E** from exposure to moisture (e.g., from defects on a top portion or an edge portion of the die **500E**).

(74) FIG. 8 schematically illustrates a top view of a die **500F** with one or more capacitor **300F** and a moat **400F**. Some of the features of the capacitor(s) **300F** and moat **400F** can be similar to any of the capacitors **300A-300E** and moat(s) **400A-400E** described in this disclosure. Thus, references numerals used to designate the various components of the capacitor(s) **300F** and moat **400F** are identical to those used for identifying the corresponding components of the capacitor(s) **300A-300E** and moat(s) **400A-400E** above, except that an “F” has been added to the numerical identifier. Therefore, the structure and description for the various features of the capacitor(s) **300A-300E** and moat(s) **400A-400E** are understood to also apply to the corresponding features of the die capacitor **300F** and moat **400F** in FIG. 8, except as described below.

(75) The die **500F** differ from the die(s) **500A-500E** described above in that the moat **400F** extends around (e.g., surrounds, circumscribes) multiple electrical components (e.g., multiple capacitor(s) **300F**). The moat **400F** can have a similar structure as any of the moat(s) **400A-400E** described above. Therefore, the moat **400F** advantageously provides a moisture barrier around at least a portion of (e.g., a majority of, substantially all of) the die **500F** to inhibit (e.g., prevent) moisture from flowing from outside the moat **400F** to the electrical components surrounded by the moat **400F**, thereby advantageously inhibiting (e.g., preventing) damage to the electrical components

(e.g., capacitor(s) **300F**) from exposure to moisture that may travel from edge defects on the die **500E**.

(76) FIG. **9** shows a method **700** of making a die with a moat in a polymer interlevel dielectric layer, such as the die **500A-500E** with moat **400A-400E**. The method **700** includes the step of forming or providing a substrate layer **710**, such as the substrate layer **510A-510E** described above. The method **700** also includes the step of forming or providing one or more metal layers **730** over the substrate layer, such as the first metal layer **530A-530E**, second metal layer **550A-550E**, third metal layer **580A-580E**, or fourth metal layer **595D**. Optionally, an insulation layer (e.g., silicon nitride layer) can be applied at least partially between metal layers, such as insulation layers **520A-520E**, **540A-540E**, **560A-560E**. The method **700** also includes forming or applying one or more polymer interlevel dielectric layers (e.g., of polyimide) **750**, such as interlayer insulator layers **570A-570E**, **590A-590E**. A trench is formed **770** in the interlayer insulator layers. Optionally, the material (e.g., polyimide) of the interlayer insulator layers is etched to form the trench. The trench can be formed in a tapered or stepped tapered manner. Advantageously, the trench in each interlayer insulator layer separates (e.g., completely separates) one side of the interlayer insulator layer from another side of the interlayer insulator layer (e.g., making the interlayer insulator layer discontinuous), thereby inhibiting (e.g., preventing) moisture from traveling across the interlayer insulator layer across the trench. The method **700** also includes the step of forming or applying a topcoat insulation layer **790**, such as topcoat insulation layer **600A-600E**, over the interlayer insulator layer(s), including over the trench. Optionally, the portion of the topcoat insulation layer that extends over the trench in the interlayer insulator layer(s) extends to the location of a layer under the interlayer insulator layer(s), such as a metal layer, an insulation layer, or the substrate layer. The topcoat insulation layer is advantageously impervious to moisture.

(77) While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

(78) Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

(79) Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a sub combination.

(80) Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in

sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

(81) For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

(82) Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

(83) Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

(84) Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

(85) The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

(86) Of course, the foregoing description is that of certain features, aspects and advantages of the present invention, to which various changes and modifications can be made without departing from the spirit and scope of the present invention. Moreover, the semiconductor die with the moisture barrier need not feature all of the objects, advantages, features and aspects discussed above. Thus,

for example, those of skill in the art will recognize that the invention can be embodied or carried out in a manner that achieves or optimizes one advantage or a group of advantages as taught herein without necessarily achieving other objects or advantages as may be taught or suggested herein. In addition, while a number of variations of the invention have been shown and described in detail, other modifications and methods of use, which are within the scope of this invention, will be readily apparent to those of skill in the art based upon this disclosure. It is contemplated that various combinations or subcombinations of these specific features and aspects of embodiments may be made and still fall within the scope of the invention. Accordingly, it should be understood that various features and aspects of the disclosed embodiments can be combined with or substituted for one another in order to form varying modes of the discussed moisture barrier in a semiconductor die.

Claims

1. A semiconductor die, comprising: a substrate layer; a capacitor over the substrate layer, the capacitor having four sides; a moat that surrounds the four sides of the capacitor to produce a moisture barrier between the capacitor and an area outside the moat, the moat includes a first trench portion that surrounds the capacitor and a second trench portion that partially surrounds the capacitor; a plurality of metal layers disposed over the substrate layer that at least partially define the capacitor, the plurality of metal layers including a first metal layer that extends under the moat on all four sides of the capacitor but is smaller in size than the substrate layer, a second metal layer over a portion of the first metal layer, the second metal layer extends under the moat on at least one side of the capacitor, but not under the moat on other sides of the capacitor; a plurality of polymer interlevel dielectric layers that partially define the capacitor, at least a first polymer interlevel dielectric layer disposed over portions of the substrate layer and under the four sides of the moat, a second polymer interlevel dielectric layer disposed over portions of the first metal layer and under the moat on at least one side of the capacitor, but not under the moat on other sides of the capacitor, the second polymer interlevel dielectric layer being tapered along the second trench portion that partially surrounds the capacitor, and a third polymer interlevel dielectric layer disposed over portions of the second metal layer, the third polymer interlevel dielectric layer being tapered along the first trench portion that surrounds the capacitor; and a topcoat insulation layer defining a top surface of a semiconductor die, the topcoat insulation layer extends over the capacitor and the moat to inhibit moisture from traveling along the plurality of polymer interlevel dielectric layers.
2. The semiconductor die of claim 1 wherein the plurality of polymer interlevel dielectric layers include one or more polyimide layers.
3. The semiconductor die of claim 1 wherein the first plurality of metal layers are one or more electrodes of the capacitor.
4. The semiconductor die of claim 1 further comprising one or more insulation layers disposed at least partially over at least one of the plurality of metal layers.
5. The semiconductor die of claim 1 wherein the first and second trench portions taper toward the substrate layer.
6. The semiconductor die of claim 1 wherein the plurality of polymer interlevel dielectric layers include at least two layers and that are stepped relative to each other.
7. The semiconductor die of claim 1 further comprising a plurality of interlayer insulation layers with a first interlayer insulation layer extending over the first metal layer and under the moat on at least one side of the capacitor, but not under the moat on three sides of the capacitor.
8. The semiconductor die of claim 1 further comprising a plurality of interlayer insulation layers with at least two interlayer insulation layers extending over the first metal layer and under the moat on at least one side of the capacitor, but not under the moat on three sides of the capacitor.
9. A radiofrequency module, comprising: a printed circuit board; and a semiconductor die mounted

on the printed circuit board and including (a) a substrate layer, (b) a capacitor over the substrate layer, the capacitor having four sides, (c) a moat that surrounds the four sides of the capacitor to produce a moisture barrier between the capacitor and an area outside the moat, the moat includes a first trench portion that surrounds the capacitor and a second trench portion that partially surrounds the capacitor; (d) a plurality of metal layers disposed over the substrate layer that at least partially define the capacitor, the plurality of metal layers including a first metal layer that extends under the moat on all four sides of the capacitor but is smaller in size than the substrate layer, a second metal layer over a portion of the first metal layer, the second metal layer extends under the moat on at least one side of the capacitor, but not under the moat on other sides of the capacitor, (e) a plurality of polymer interlevel dielectric layers that partially define the capacitor, at least a first polymer interlevel dielectric layer disposed over portions of the substrate layer and under the four sides of the moat, and a second polymer interlevel dielectric layer disposed over portions of the first metal layer and under the moat on at least one side of the capacitor, but not under the moat on other sides of the capacitor, the second polymer interlevel dielectric layer being tapered along at least one side of the second trench portion that partially surrounds the capacitor, and a third polymer interlevel dielectric layer disposed over portions of the second metal layer, the third polymer interlevel dielectric layer being tapered along the first trench portion that surrounds the capacitor, and (f) a topcoat insulation layer disposed over defining a top surface of the semiconductor die, the topcoat insulation layer extends over the capacitor and the moat to inhibit moisture from traveling along the plurality of polymer interlevel dielectric layers.

10. The radiofrequency module of claim 9 wherein the plurality of polymer interlevel dielectric layers are one or more polyimide layers.

11. The radiofrequency module of claim 9 wherein the first plurality of metal layers are one or more electrodes of the capacitor.

12. The radiofrequency module of claim 9 further comprising one or more insulation layers disposed at least partially over at least one of the plurality of metal layers.

13. The radiofrequency module of claim 9 wherein the plurality of polymer interlevel dielectric layers include at least two layers that are stepped relative to each other.

14. The radiofrequency module of claim 9 further comprising a plurality of interlayer insulation layers with a first interlayer insulation layer extending over the first metal layer and under the moat on at least one side of the capacitor, but not under the moat on three sides of the capacitor.

15. A wireless mobile device, comprising: one or more antennas; a front end system that communicates with the one or more antennas; and a semiconductor die including (a) a substrate layer, (b) a capacitor over the substrate layer, the capacitor having four sides, (c) a moat that surrounds the four sides of the capacitor to produce a moisture barrier between the capacitor and an area outside the moat, the moat includes a first trench portion that surrounds the capacitor and a second trench portion that partially surrounds the capacitor; (d) a plurality of metal layers disposed over the substrate layer that at least partially define the capacitor, the plurality of metal layers including a first metal layer that extends under the moat on all four sides of the capacitor but is smaller in size than the substrate layer, a second metal layer over a portion of the first metal layer, the second metal layer extends under the moat on at least one side of the capacitor, but not under the moat on other sides of the capacitor, (e) a plurality of polymer interlevel dielectric layers that partially define the capacitor, at least a first polymer interlevel dielectric layer disposed over the substrate layer and under the four sides of the moat, a second polymer interlevel dielectric layer disposed over portions of the first metal layer and under the moat on at least one side of the capacitor, but not on other sides of the capacitor, the second polymer interlevel dielectric layer being tapered along the second trench portion that partially surrounds the capacitor, and a third polymer interlevel dielectric layer disposed over portions of the second metal layer, the third polymer interlevel dielectric layer being tapered along the first trench portion that surrounds the capacitor, and (f) a topcoat insulation layer defining a top surface of the semiconductor die, the

topcoat insulation layer extends over the capacitor and the moat to inhibit moisture from traveling along the plurality of polymer interlevel dielectric layers.

16. The wireless mobile device of claim 15 wherein the plurality of polymer interlevel dielectric layers include one or more polyimide layers.

17. The wireless mobile device of claim 15 wherein the first plurality of metal layers are one or more electrodes of the capacitor.

18. The wireless mobile device of claim 15 further comprising one or more insulation layers disposed at least partially over at least one of the plurality of metal layers.

19. The wireless mobile device of claim 15 wherein the plurality of polymer interlevel dielectric layers include at least two layers that are stepped relative to each other.

20. The wireless mobile device of claim 15 further comprising a plurality of interlayer insulation layers with a first interlayer insulation layer extending over the first metal layer and under the moat on least one side of the capacitor, but not under the moat on three sides of the capacitor.
